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University of Moratuwa, Sri Lanka

BM4152 Biosignal Processing

Assignment 02

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1. Wiener Filter

1.1. Data Construction

Fig. 1 shows the ideal and constructed noisy ECG signals.

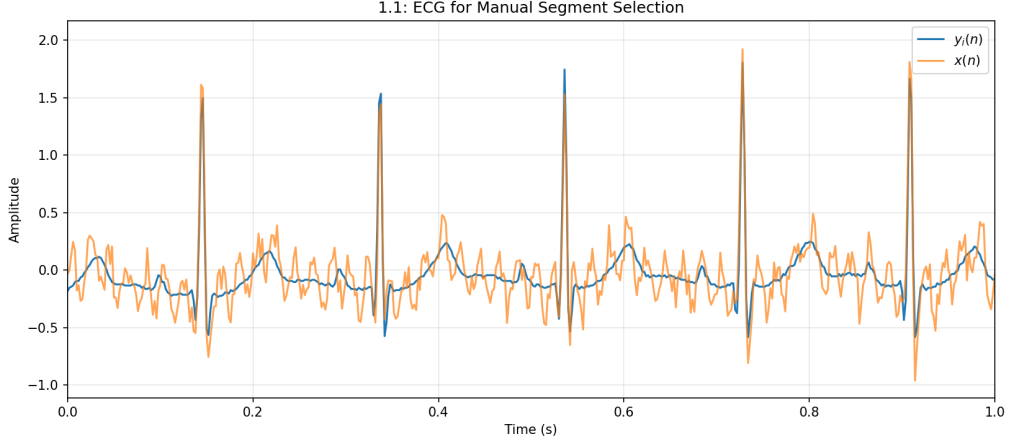


Figure 1: Ideal ECG signal and noisy input signal.

For the Wiener filter calculation, one ECG beat was selected from the ideal signal and a noise reference was selected from an isoelectric segment in the noisy signal. The selected regions are shown in Figs. 2 and 3.

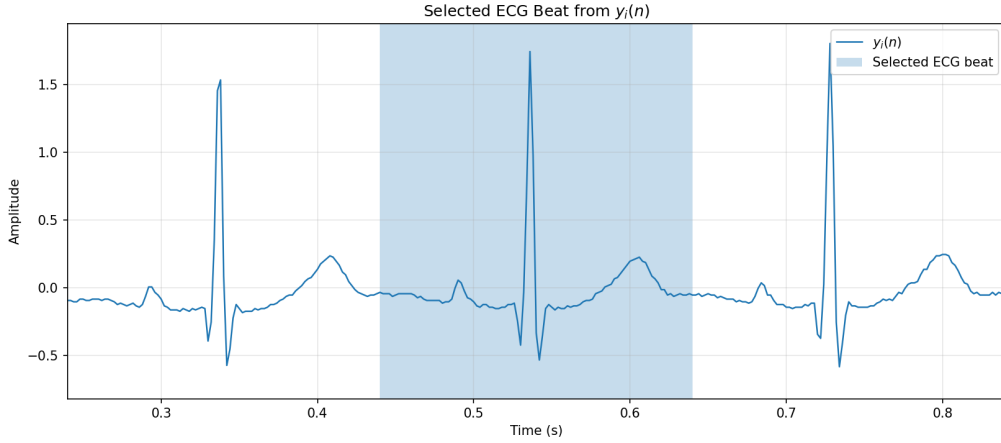


Figure 2: Selected ECG beat from the ideal signal.

1.2. Discrete Time-Domain Wiener Filter

1.2.1 Part 1: Ideal ECG Beat

a. Arbitrary Filter Order

An arbitrary number of coefficients of $M = 10$ was initially selected. Therefore, the corresponding FIR filter order was $M - 1 = 9$. The resulting Wiener filter coefficients were

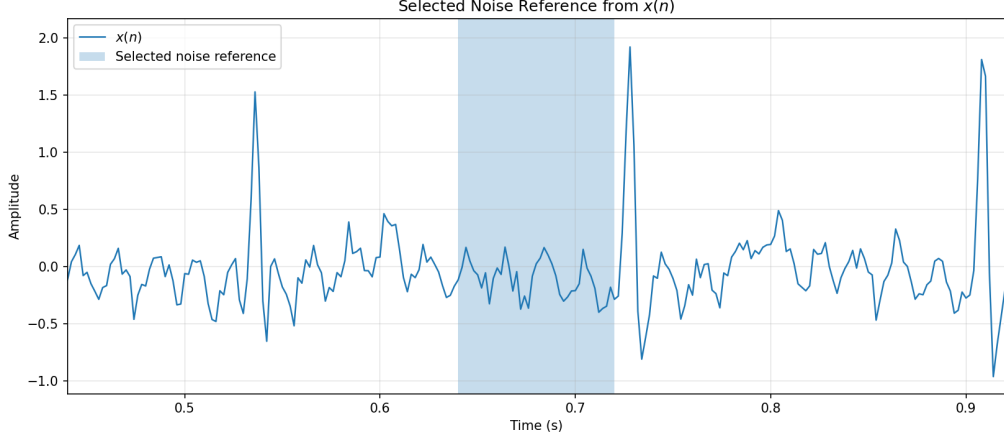


Figure 3: Selected noise reference from the isoelectric segment of the noisy ECG.

$$\mathbf{w}_0 = \begin{bmatrix} 0.56825 & 0.18298 & -0.15361 & -0.10528 & 0.10299 \\ -0.02906 & -0.01314 & 0.02648 & -0.02697 & -0.11488 \end{bmatrix}. \quad (1)$$

The MSE between the filtered ECG beat and the ideal ECG beat was $\text{MSE} = 0.022094$.

The noisy and filtered ECG signals are shown in Fig. 4.

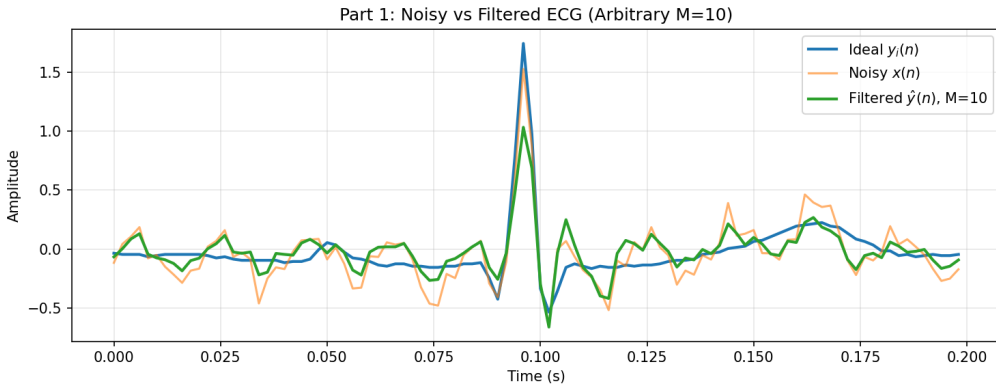


Figure 4: Noisy and filtered ECG using the arbitrary filter order.

b. Optimum Filter Order

The filter order was varied by changing the number of coefficients from $M = 2$ to $M = 40$. The minimum MSE was obtained for $M = 21$, corresponding to an FIR filter order of 20. The minimum MSE obtained during the filter-order search was $\text{MSE}_{\min} = 0.021574$. The variation of MSE with the number of filter coefficients is shown in Fig. 5.

The figure 6 shows the magnitude response of the optimal filter,

c. Filtered signal (Optimal)

The optimum filter with $M = 21$ was then applied to the selected noisy ECG beat. The resulting MSE was $\text{MSE} = 0.018986$.

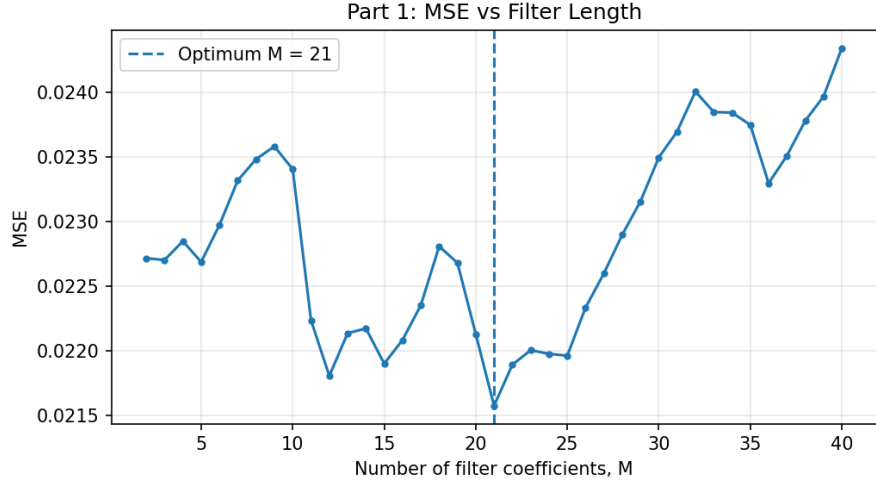


Figure 5: Part 1: MSE as a function of the number of Wiener filter coefficients.

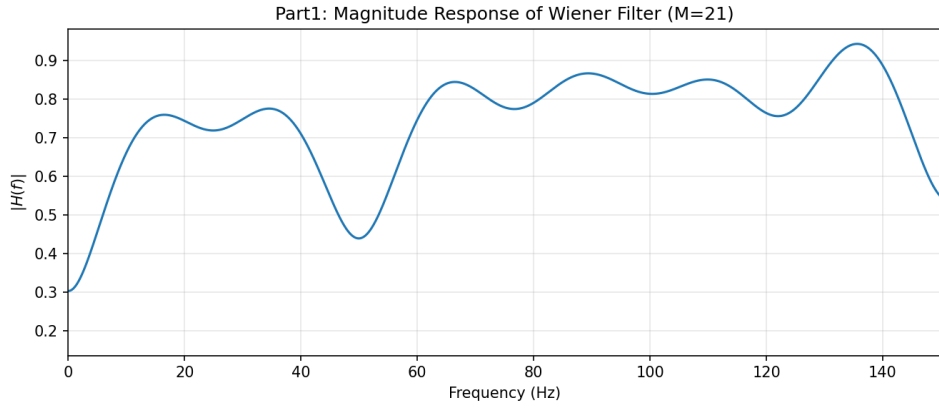


Figure 6: Magnitude response of the optimum Wiener filter

The ideal, noisy, and filtered signals are shown in Fig. 7.

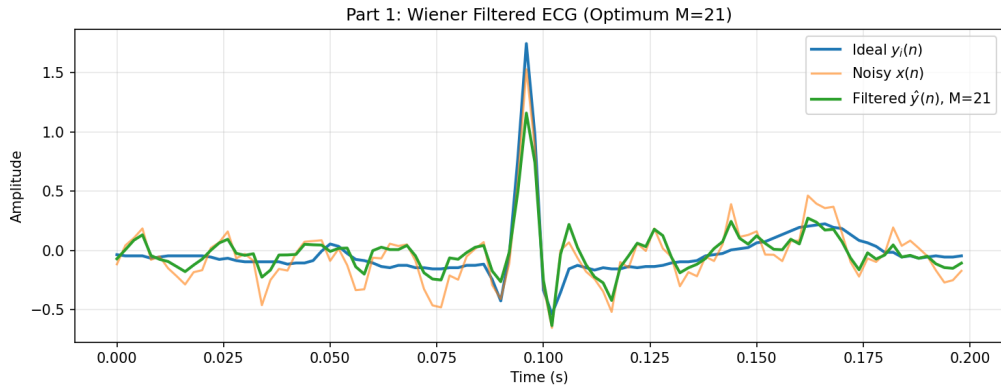


Figure 7: Part 1: ECG filtered using the optimum Wiener filter.

d. Spectral Analysis

The power spectral densities of the ideal ECG, noise, noisy ECG, and filtered ECG were calculated using a periodogram.

The spectra of the four signals are shown in Fig. 8.

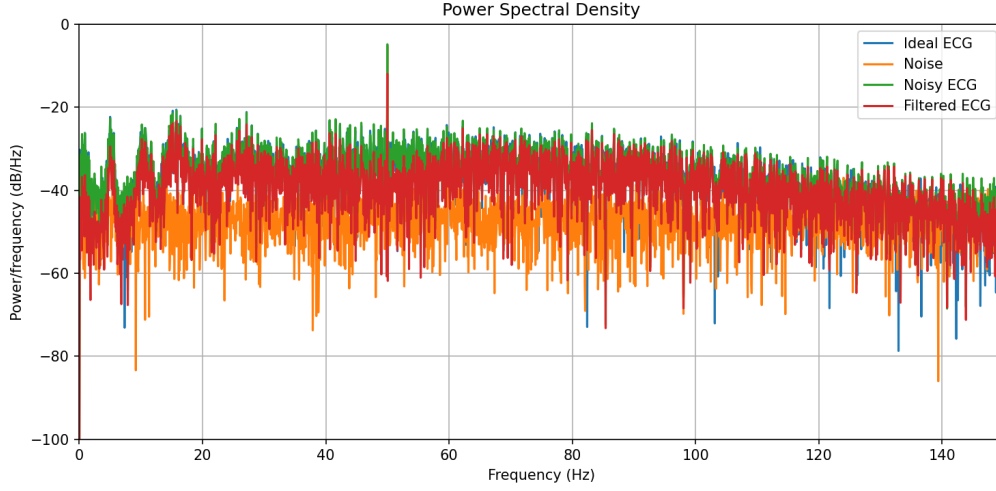


Figure 8: Power spectral densities for Part 1.

e. Interpretation

The spectrum of the noisy ECG contains a strong component around 50 Hz due to the sinusoidal power-line interference. After Wiener filtering, the 50 Hz component is reduced considerably while retaining the main low-frequency components of the ECG. This shows that the Wiener filter can suppress the dominant interference while retaining the frequency components associated with the ECG signal.

The magnitude response shows an attenuation around 50 Hz; i.e the filter has relatively high gain in frequency regions where the ECG has more power and lower gain where the noise dominates.

1.2.2 Part 2: Constructed Linear ECG Model

For Part 2, a piecewise-linear ECG model was constructed with the same length as the selected ECG beat. The constructed model is shown in Fig. 9.

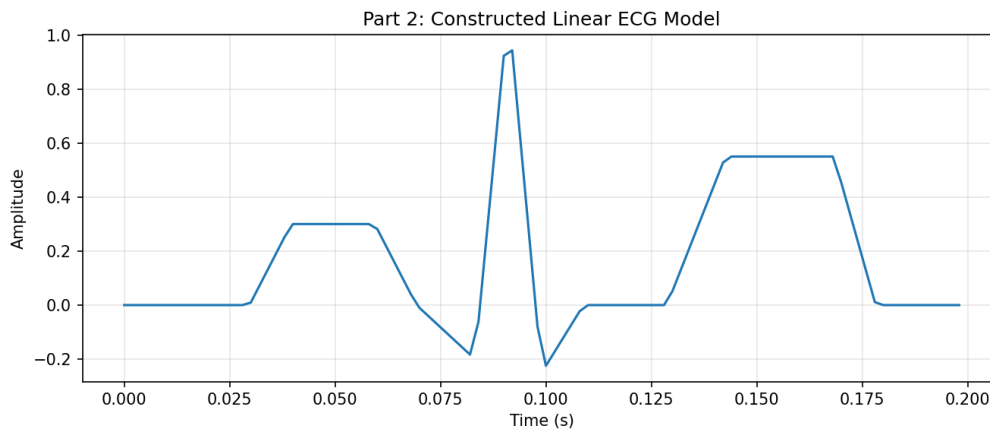


Figure 9: Constructed piecewise-linear ECG model used as the desired signal in Part 2.

a. Arbitrary Filter Order

An arbitrary number of coefficients of $M = 10$ was first selected, giving an FIR filter order of 9. The MSE obtained by applying this filter was $\text{MSE} = 0.021710$. The corresponding filtered signal is shown in Fig. 10.

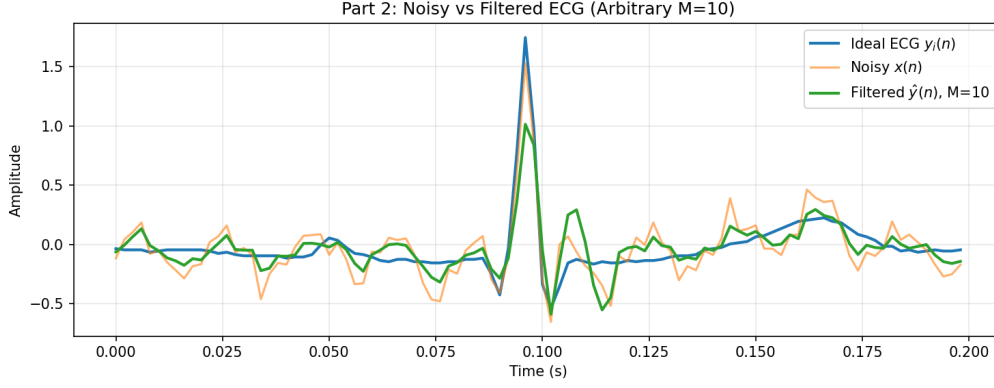


Figure 10: Part 2: Noisy and filtered ECG using the arbitrary filter order.

b. Optimum Filter Order

The filter length was varied from $M = 2$ to $M = 40$. The minimum MSE during the filter-order search was obtained at $M = 15$ corresponding to an FIR filter order of 14. The minimum MSE obtained during the search was $\text{MSE}_{\min} = 0.021304$. The MSE variation with filter length is shown in Fig. 11.

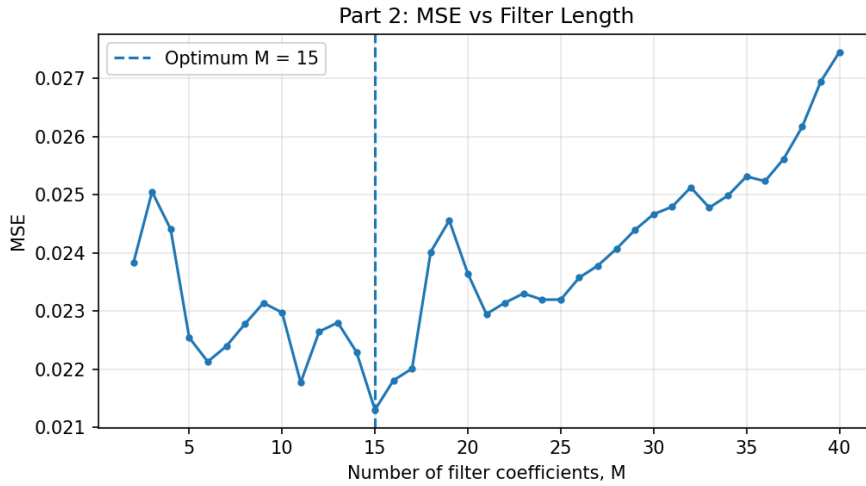


Figure 11: Part 2: MSE as a function of the number of Wiener filter coefficients.

The figure 12 shows the magnitude response of the optimal filter,

c. Filtered signal (Optimal)

Using the optimum filter with $M = 15$, the resulting MSE was $\text{MSE} = 0.019522$. The resulting filtered ECG signal is shown in Fig. 13.

d. Spectral Analysis

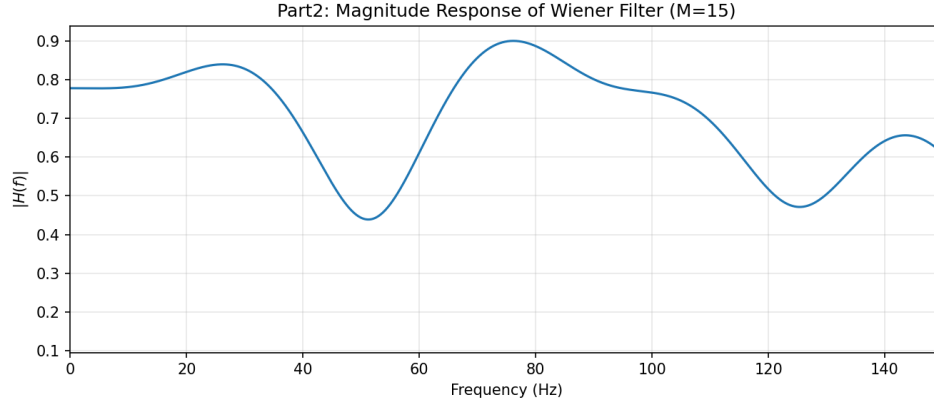


Figure 12: Magnitude response of the optimum Wiener filter for Part 2.

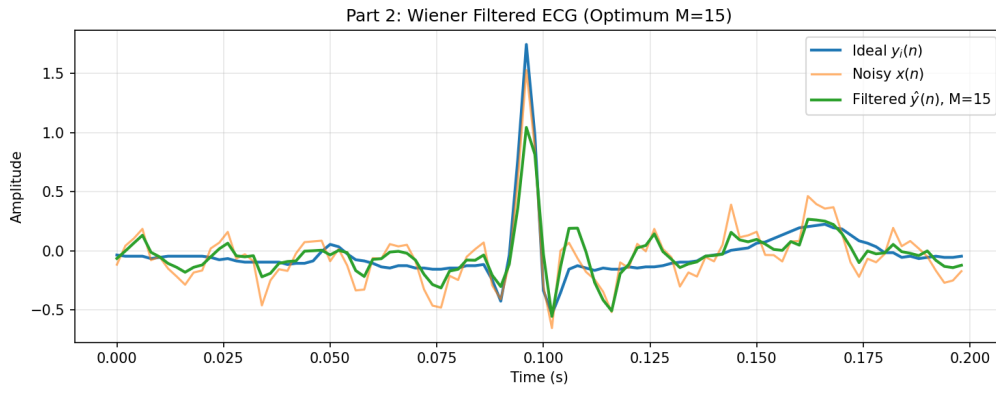


Figure 13: Part 2: ECG filtered using the optimum Wiener filter.

The optimum Part 2 Wiener filter was applied to the full noisy ECG signal. The PSD comparison is shown in Fig. 14.

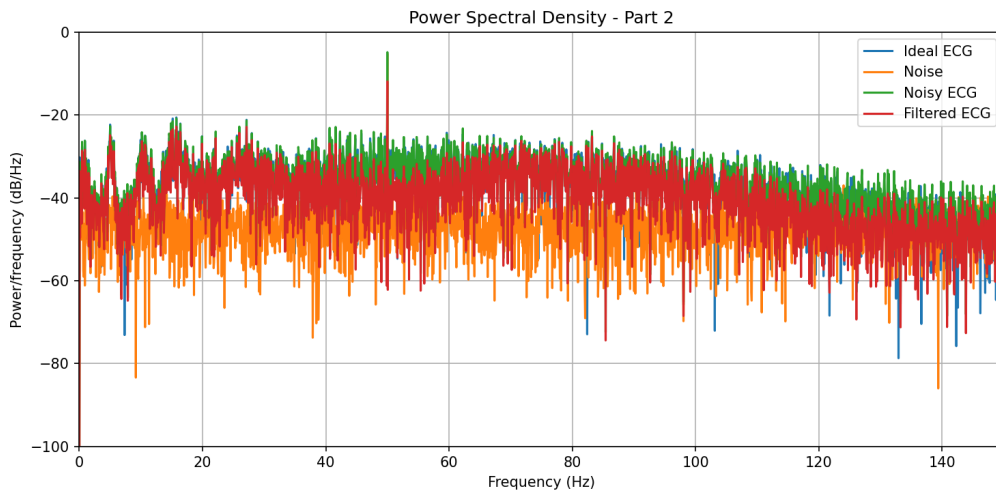


Figure 14: Power spectral densities for Part 2.

e. Interpretation

Similar to Part 1, the noisy ECG contains a strong 50 Hz component. The Wiener filter reduces this component while preserving the main frequency components of the ECG.

1.3. Frequency-Domain Implementation of the Wiener Filter

The frequency-domain Wiener filter was implemented using

$$W(f) = \frac{S_{YY}(f)}{S_{YY}(f) + S_{NN}(f)}, \quad (2)$$

where the signal and noise power spectra were calculated using

$$S_{ZZ}(f) = |\mathcal{F}\{z(n)\}|^2. \quad (3)$$

The noisy ECG was then filtered in the frequency domain according to

$$\hat{Y}(f) = W(f)X(f). \quad (4)$$

The filtered signal was obtained by applying the inverse Fourier transform:

$$\hat{y}(n) = \mathcal{F}^{-1}\{\hat{Y}(f)\}. \quad (5)$$

The MSE values obtained for the two parts are summarized in Table 1.

Table 1: MSE of the Frequency-Domain Wiener Filter

Part	MSE
Part 1	0.015324
Part 2	0.017906

The filtered signals obtained from the two cases are shown in Figs. 15 and 16.

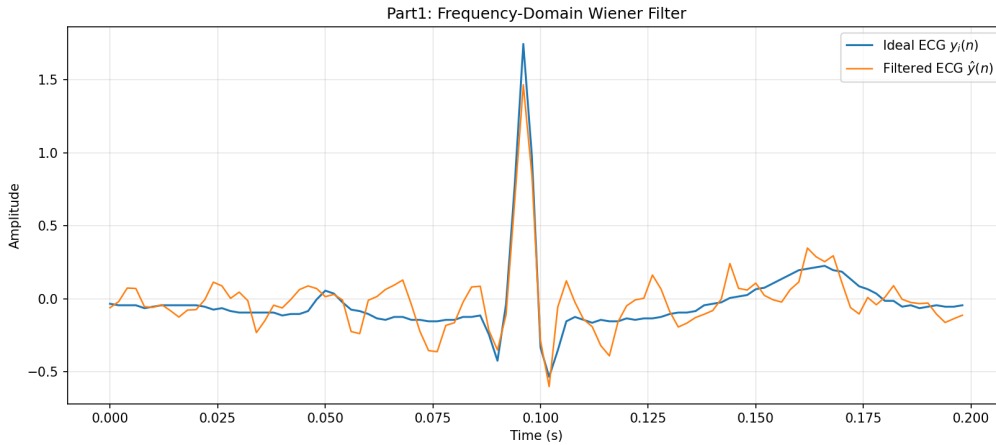


Figure 15: Part 1: Frequency-domain Wiener-filtered ECG.

The frequency-domain Wiener filter reduced the noise in both cases while preserving the main morphology of the ECG signal. The MSE values differed slightly because the desired signal used to construct the Wiener filter differed between the two cases.

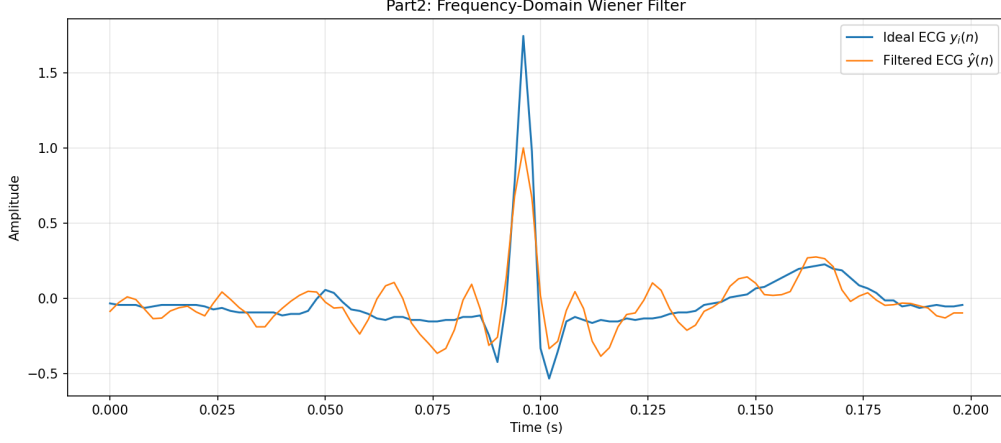


Figure 16: Part 2: Frequency-domain Wiener-filtered ECG.

1.4. Effect of Non-Stationary Noise on Wiener Filtering

The 50 Hz sinusoidal interference was changed to 100 Hz halfway through the signal duration. The noise was therefore defined as

$$\eta(n) = \begin{cases} \eta_{wg}(n) + 0.2 \sin(2\pi 50t), & 0 \leq n < T/2, \\ \eta_{wg}(n) + 0.2 \sin(2\pi 100t), & T/2 \leq n < T. \end{cases} \quad (6)$$

The optimum Part 1 Wiener filter was applied to this new signal.

Fig. 17 shows the resulting signal.

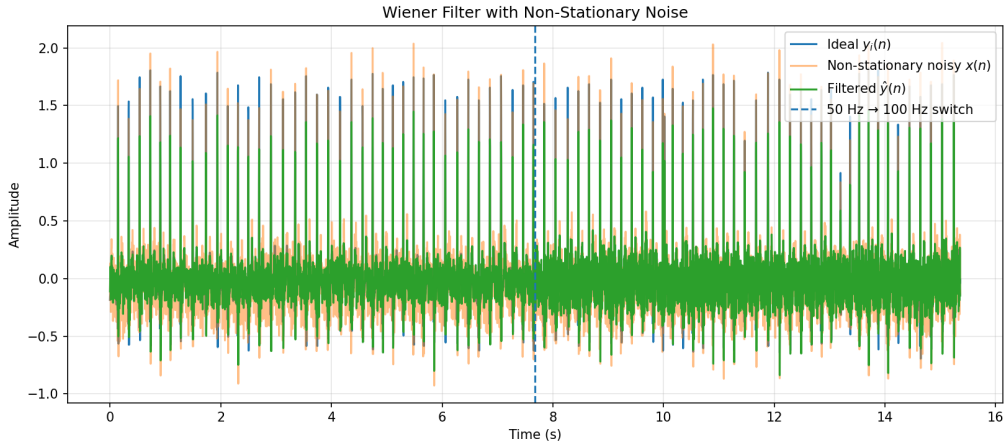


Figure 17: Fixed Wiener filter applied to the ECG with non-stationary noise.

A zoomed view around the frequency-switching point is shown in Fig. 18.

Interpretation: The Wiener filter used in this experiment was designed using the original noise characteristics, which included the 50 Hz interference. Therefore, it is better matched to the noise in the first half of the signal. After the interference changes from 50 Hz to 100 Hz at $T/2$, the noise characteristics are no longer the same as those used to design the filter. As we can see, more residual interference remains in the second half of

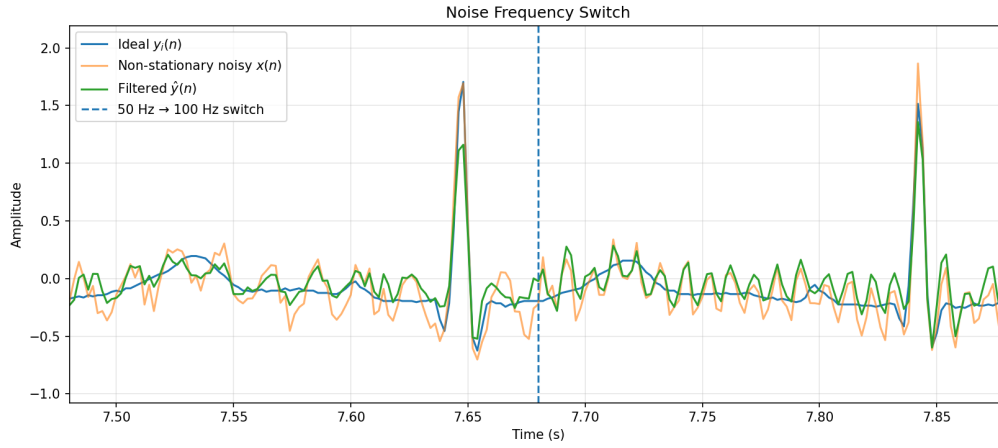


Figure 18: Zoomed view around the 50 Hz to 100 Hz noise-frequency switch.

the filtered signal. This demonstrates that **a fixed Wiener filter is less suitable for non-stationary noise.**

2. Adaptive Filtering

2.1. Data Construction

The primary input signal was constructed as

$$x(n) = y_i(n) + \eta(n), \quad (7)$$

where the desired signal $y_i(n)$ was a sawtooth waveform with a width of 0.5 and $\eta(n)$ consists of white Gaussian noise and a non-stationary sinusoidal interference.

The reference signal was constructed according to

$$r(n) = a [\eta_{wg}(n) + \sin(2\pi 50t + \phi_1) + \sin(2\pi 100t + \phi_2)], \quad (8)$$

where $a = 0.5$, $\phi_1 = \pi/4$, and $\phi_2 = \pi/3$. The same white Gaussian noise realization used in $\eta(n)$ was also included in the reference signal. A fundamental frequency of $1Hz$ was chosen for the sawtooth so that a handful of periods are visible over the 10 s record

2.2. LMS Method

2.2.1 Implementation

The Least Mean Squares (LMS) adaptive filter was implemented according to

$$e(n) = x(n) - \mathbf{w}^T(n)\mathbf{R}(n), \quad (9)$$

where

$$\mathbf{R}(n) = [r(n), r(n-1), \dots, r(n-M+1)]^T. \quad (10)$$

The adaptive weights were updated using

$$\mathbf{w}(n+1) = \mathbf{w}(n) + 2\mu e(n)\mathbf{R}(n). \quad (11)$$

The error signal $e(n)$ is used as the estimate $\hat{y}(n)$ of the desired signal $y_i(n)$.

For the initial LMS experiment, the step size was set to $\mu = 0.01$ and the number of filter taps was set to $M = 15$.

2.2.2 LMS Filtering Result

Figure 19 shows the desired sawtooth signal, the noisy primary input, the LMS output, and the absolute error.

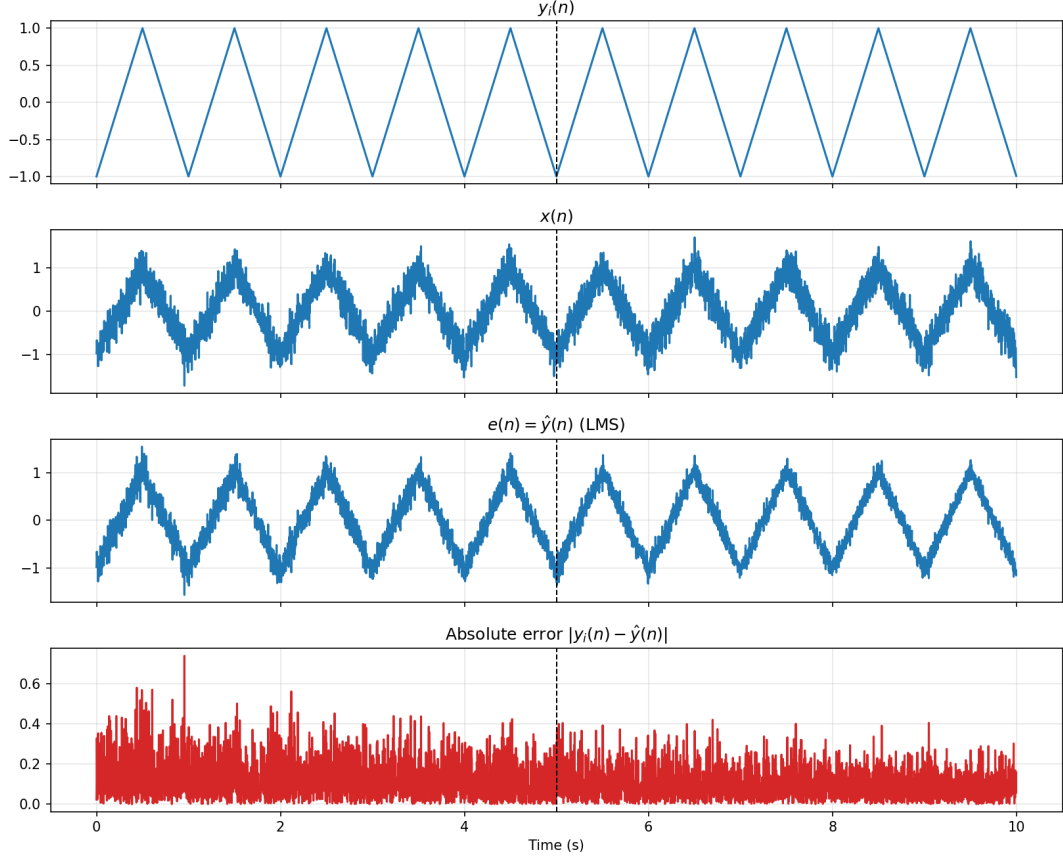


Figure 19: LMS filtering result for the sawtooth test signal.

The mean squared error was calculated over the complete signal using

$$\text{MSE} = \frac{1}{N} \sum_{n=0}^{N-1} (y_i(n) - \hat{y}(n))^2. \quad (12)$$

For $\mu = 0.01$ and $M = 15$, the obtained MSE was $\text{MSE}_{\text{LMS}} = 0.02078$.

2.2.3 Effect of Step Size and Filter Order

The optimal parameters of the LMS algorithm were obtained by varying the step size μ and the number of filter taps M . The step size was varied from 0.001 to 0.05 using 20 equally spaced values, while the number of taps was varied from 1 to 30. The resulting MSE values are shown in the heatmap in Figure 20.

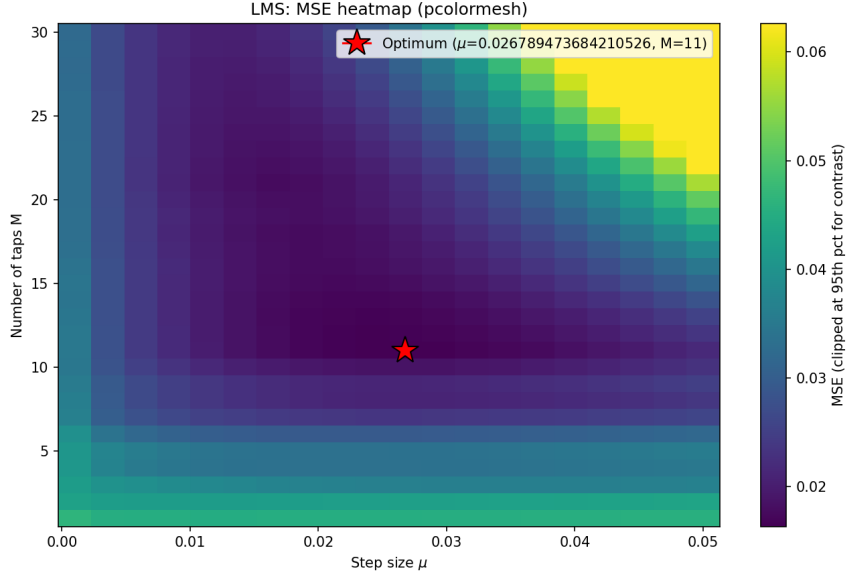


Figure 20: LMS MSE for different step sizes μ and numbers of filter taps M .

The minimum MSE was obtained at $\mu = 0.02679$, $M = 11$, corresponding to a filter order $(M - 1) = 10$, with $\text{MSE} = 0.01633$.

The MSE values show a noticeable reduction with the selected parameters, indicating a better adaptation rate of the LMS filter to the changing noise.

The LMS filter result using the optimum parameters is shown in Figure 21.

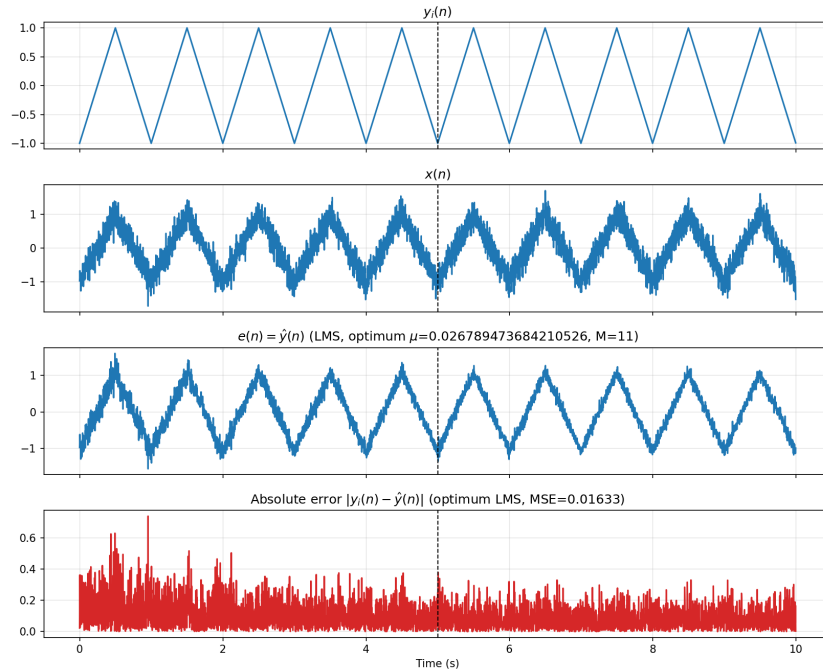


Figure 21: LMS filtering result using the optimal values obtained

2.3. RLS Method

2.3.1 Implementation

The Recursive Least Squares (RLS) adaptive filter was implemented using the equations given. The gain vector was calculated as

$$\mathbf{k}(n) = \frac{\lambda^{-1}\mathbf{P}(n-1)\mathbf{R}(n)}{1 + \lambda^{-1}\mathbf{R}^T(n)\mathbf{P}(n-1)\mathbf{R}(n)}. \quad (13)$$

The inverse correlation matrix was updated using

$$\mathbf{P}(n) = \lambda^{-1}\mathbf{P}(n-1) - \lambda^{-1}\mathbf{k}(n)\mathbf{R}^T(n)\mathbf{P}(n-1). \quad (14)$$

The a priori error was calculated as

$$\alpha(n) = x(n) - \mathbf{w}^T(n-1)\mathbf{R}(n), \quad (15)$$

and the adaptive weights were updated according to

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \mathbf{k}(n)\alpha(n). \quad (16)$$

The output of the adaptive filter was then calculated as

$$\hat{y}(n) = x(n) - \mathbf{w}^T(n)\mathbf{R}(n). \quad (17)$$

With the initial conditions $\mathbf{P}(0) = \delta^{-1}\mathbf{I}$, $\mathbf{w}(0) = \mathbf{0}$

2.3.2 Comparison of LMS and RLS

For a direct comparison, both algorithms were evaluated using the same filter length of $M = 15$. The LMS step size was $\mu = 0.01$, while the RLS forgetting factor was selected as $\lambda = 0.995$.

The results are summarized in Table 2.

Table 2: LMS and RLS performance for the sawtooth signal.

Method	Parameter	Taps M	MSE
LMS	$\mu = 0.01$	15	0.02078
RLS	$\lambda = 0.995$	15	0.02542

The resulting outputs and absolute errors are compared in Figure 22.

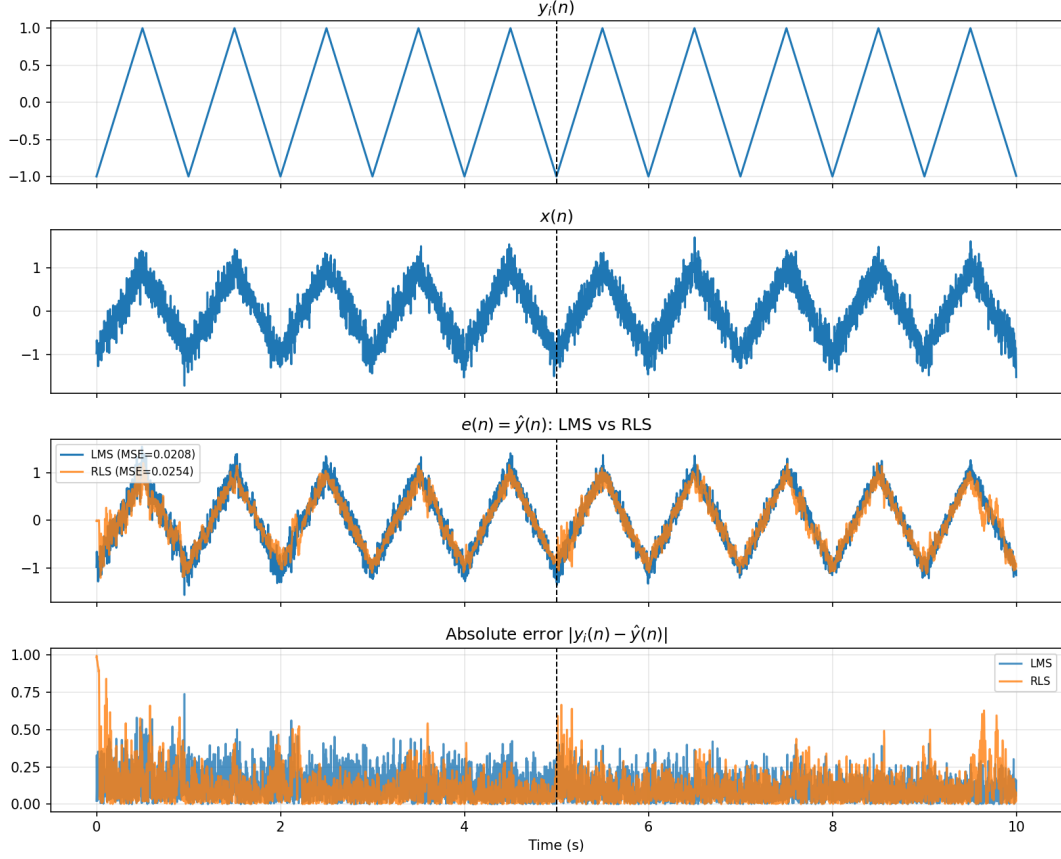


Figure 22: Comparison of LMS and RLS adaptive filtering for the sawtooth signal.

It should be noted that the comparison is dependent on the selected adaptation parameters (μ and λ). Therefore, we cannot conclude which algorithm performed better based on selecting the same filter length.

2.3.3 Effect of Forgetting Factor and Filter Order

The RLS adaptation behaviour was investigated by varying the forgetting factor λ and the number of filter taps M . The forgetting factor was varied from 0.90 to 1.00 using 21 equally spaced values, while the number of taps was varied from 1 to 30.

The resulting MSE values are shown in Figure 23.

The minimum MSE obtained at $\lambda = 0.995, M = 7$, with $MSE = 0.02175$. The corresponding filter order is $M - 1 = 6$

The heatmap shows that the RLS performance changes with both the forgetting factor and filter length. The results also show that choosing suitable values for both parameters can reduce the MSE and improve the adaptation of the filter.

The RLS filter result using the optimum parameters is shown in Figure 24.

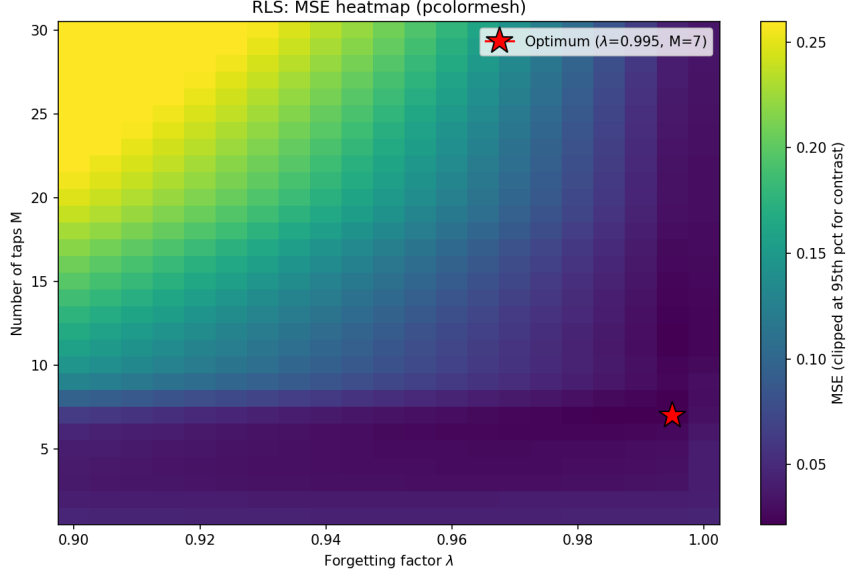


Figure 23: RLS MSE for different forgetting factors λ and numbers of filter taps M .

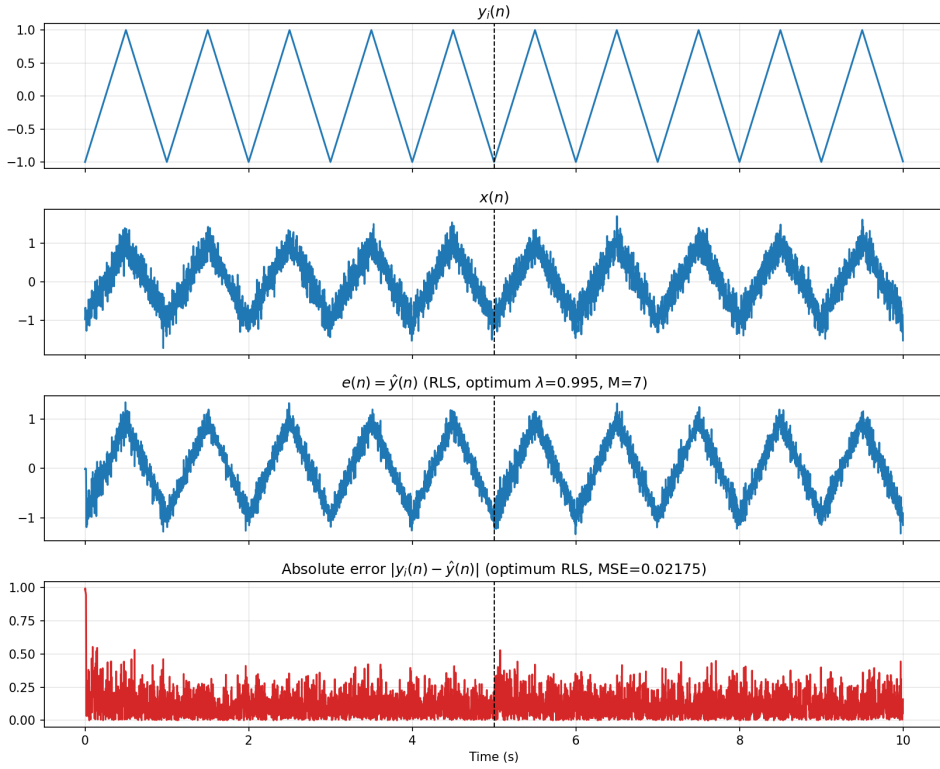


Figure 24: RLS filtering result using the optimal parameters obtained.

2.4. Evaluation Using the idealECG Signal

The LMS and RLS algorithms were tested using the `idealECG` signal as the desired signal. The same noise generation procedure was retained, while $y_i(n)$ was replaced by the ECG signal.

The algorithms were evaluated using the same values used in 2.2.a) $\mu = 0.01, M = 15$ for

LMS and $\lambda = 0.995$, $M = 15$ for RLS.

Table 3: LMS and RLS performance using the idealECG signal.

Method	MSE
LMS	0.01122
RLS	0.00502

The corresponding filtering results are shown in Figure 25.

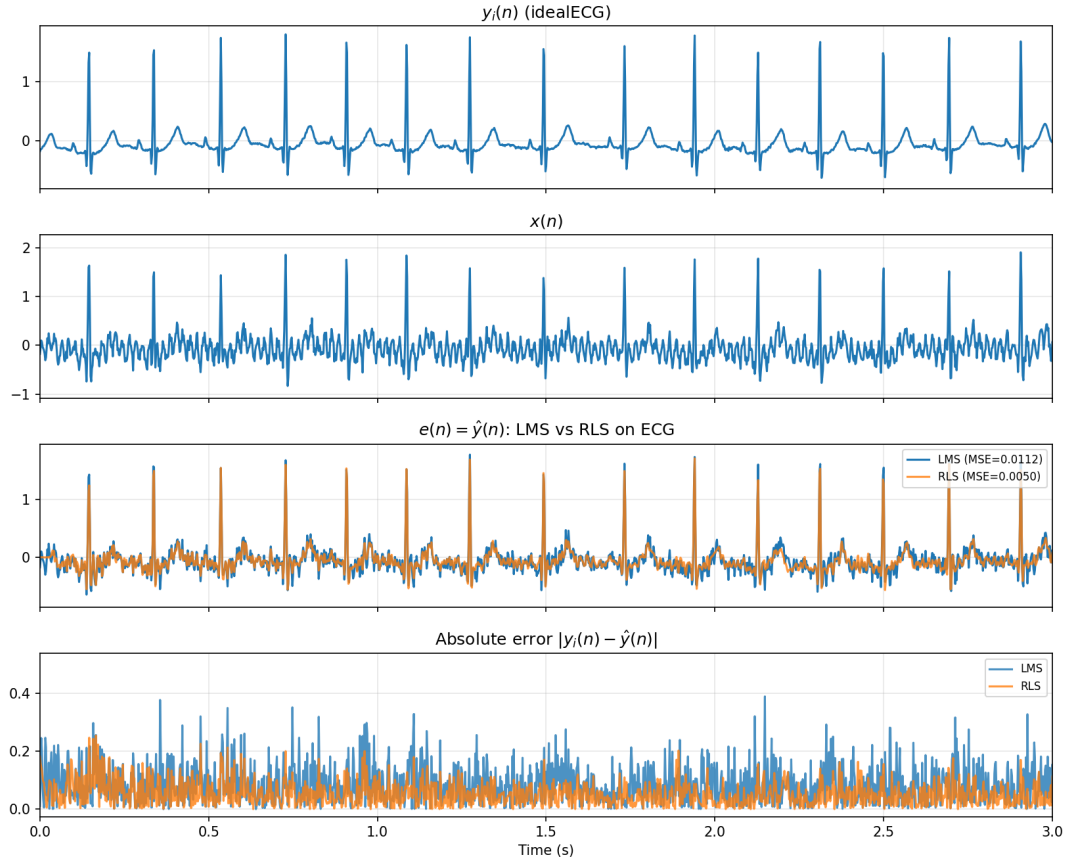


Figure 25: Implementation of LMS and RLS adaptive filters using the idealECG signal.

The ECG experiment produced different MSE values(very low) from the sawtooth experiment because the desired signals have different characteristics. This shows that the performance of an adaptive filtering algorithm depends on the characteristics of the input signals and the selected adaptation parameters.